

Below is the CESD Laravel/Next.js system blueprint.

CESD Website & Service Platform

1. Technology Stack

Frontend: Next.js
Backend/API: Laravel
Admin Panel: Laravel Filament
Database: PostgreSQL or MySQL
Authentication: Laravel Sanctum
Languages: Lao / English
Storage: Local server or S3-compatible storage

CESD should be designed as a bilingual public website plus service-management platform for laboratories, equipment, testing, training, and industry collaboration.

2. Main User Roles

1. Public Visitor
2. Customer / Industry Partner
3. Student / Researcher
4. Laboratory Staff
5. Unit Head
6. CESD Director / Deputy Director
7. System Administrator

The organization structure should support the Director, Deputy Director, Administration and Planning Unit, Technical Service and Laboratory Unit, and Innovation/AI Database/Training Unit, as proposed in the CESD establishment document.

3. Website Sitemap

Home

About CESD

- Background
- Vision & Mission
- Objectives
- Organization Structure

Laboratories

- Chemical Analysis Laboratory
- Electron Microscope Laboratory
- X-ray Analysis Laboratory
- High-Voltage Laboratory
- Solid Forming Laboratory
- Computer Room
- Physical Measurement Laboratory
- Electrical and Electronic Measurement Laboratory

Equipment Catalog

Services

- Testing & Analysis
- Product Inspection
- Performance Test
- Joint R&D
- Technical Consulting
- Training / Workshop
- Facility Booking

Projects & Partners

Training

News & Events

Contact

Request Service

Login / Dashboard

The laboratory list should follow the CESD presentation, which identifies chemical analysis, electron microscope, X-ray analysis, high-voltage, solid forming, computer room, physical measurement, electrical/electronic measurement, meeting room, exhibition room, and administration office.

4. Core Functional Modules

Module A: Bilingual Content Management

- Manage Lao and English content
- Pages: About, Vision, Mission, Services, Laboratories
- News and events
- Projects and partners
- Banner and homepage highlights

Module B: Laboratory Management

- Add/edit laboratory
- Assign responsible staff
- Add location, photos, safety rules
- Link laboratory to equipment
- Show laboratory profile publicly

Module C: Equipment Catalog

- Equipment name Lao / English
- Equipment code
- Laboratory
- Category
- Brand/model
- Testing capability
- Technical specification
- Availability status
- Responsible staff

- User manual / safety document
- Calibration and maintenance records

The CESD documents mention more than 300 laboratory equipment items supported by JICA, so the catalog must support many records and search/filter functions.

Module D: Testing / Analysis Service Request

Customer submits request
Admin reviews request
Lab staff evaluates feasibility
Quotation is prepared
Customer confirms
Sample is submitted
Testing is performed
Result is reviewed
Report is issued
Customer downloads report

CESD's intended services include product inspection, performance tests, measurement, analysis, and external technical services.

Module E: Equipment / Facility Booking

User logs in
Searches equipment or room
Checks available schedule
Submits booking
Lab staff approves/rejects
User receives confirmation
Usage record is stored

Module F: Training / Workshop Management

- Training course listing
- Registration form
- Participant management
- Attendance
- Certificate generation
- Corporate training request

This supports CESD's role in talent development, customized training, and workshops for industry.

Module G: Partnership / Joint R&D

- Partnership inquiry
- Partner profile
- MoU tracking
- Joint project record
- Project outputs

- Publication/showcase

Module H: CRM and Report Verification

- Customer profiles
- Service history
- Request tracking
- QR-code report verification
- Feedback collection

The Lao CESD document specifically recommends CRM linked with database/AI functions and QR-code verification for reports.

5. Proposed Database Tables

users

roles

permissions

user_roles

pages

page_translations

news

news_translations

projects

project_translations

laboratories

laboratory_translations

equipment

equipment_translations

equipment_categories

equipment_documents

equipment_maintenance

equipment_calibrations

services

service_translations

service_requests

service_request_items

service_status_logs

quotations

quotation_items

payments

bookings

booking_status_logs

training_courses

training_translations

training_registrations

certificates

partners

partnership_requests
joint_projects

reports
report_files
report_verifications

contacts
feedback
settings
audit_logs

6. Important Database Fields

laboratories

id
code
name_en
name_lo
description_en
description_lo
location
floor
responsible_user_id
status
created_at
updated_at

equipment

id
laboratory_id
category_id
code
name_en
name_lo
brand
model
serial_number
specification_en
specification_lo
capability_en
capability_lo
status
availability_status
photo
manual_file
created_at
updated_at

service_requests

id
request_no
customer_id
service_type
laboratory_id
equipment_id
title
description
sample_information
required_date
status
assigned_staff_id
quotation_id
created_at
updated_at

bookings

id
booking_no
user_id
equipment_id
laboratory_id
room_id
start_datetime
end_datetime
purpose
status
approved_by
approved_at
created_at
updated_at

reports

id
service_request_id
report_no
title
result_summary
file_path
qr_code
verification_token
approved_by
approved_at
issued_at
status
created_at
updated_at

7. Development Phases

Phase 1: Public Website + Admin CMS

- Lao/English website
- About CESD
- Vision/Mission/Objectives
- Laboratory directory
- Equipment catalog
- Services page
- News/events
- Contact page
- Admin dashboard

Phase 2: Service Platform

- Online service request
- Equipment/facility booking
- Customer dashboard
- Staff workflow
- Quotation management
- Status tracking

Phase 3: Advanced LIMS Functions

- Digital reports
- QR-code verification
- CRM
- Calibration and maintenance
- Analytics dashboard
- AI/search recommendation

8. Developer Prompt

Build a bilingual Lao/English website and service-management platform for CESD: Center of Engineering for Sustainable Development, Faculty of Engineering, National University of Laos.

Use Next.js for the frontend, Laravel for backend API, Laravel Filament for admin dashboard, Laravel Sanctum for authentication, and PostgreSQL/MySQL for database.

The system must include:

1. Public bilingual website
2. Laboratory directory
3. Equipment catalog with search/filter
4. Testing and analysis service request workflow
5. Equipment and facility booking workflow
6. Training/workshop management
7. Partnership and joint R&D inquiry
8. Customer CRM
9. Quotation and service status tracking
10. Digital report upload/download with QR-code verification
11. Admin dashboard with role-based access control

The CESD laboratories include:

- Chemical Analysis Laboratory
- Electron Microscope Laboratory

- X-ray Analysis Laboratory
- High-Voltage Laboratory
- Solid Forming Laboratory
- Computer Room
- Physical Measurement Laboratory
- Electrical and Electronic Measurement Laboratory
- Meeting Room
- Exhibition Room
- Administration Office

The platform must support CESD services for student project support, curriculum integration, faculty research, contracted analysis, joint R&D, talent development, customized training, facility sharing, and external industry service.

This is ready to become a **Software Requirement Specification (SRS)** or a **developer task package**.